Homework 3: solutions

Expectation, variance, and joint discrete laws

These solutions cover all ten questions. Mathematically equivalent methods, exact expressions, and complete alternative PMF notation are valid. When a question asks for two calculations, both are shown. Questions 4 and 7 are worth 10 points each; the other questions are worth 5 points each, for a total of 60.

1. Expectation of a discrete random variable

5 points

A random variable X has the following PMF, with zero mass elsewhere:

x	-3	1	4
$p_X(x)$	$1/4$	$1/2$	$1/4$

(a) [2] Compute $E[X]$ from the definition of expectation. Can X take the value $E[X]$? Explain how your answer relates to the meaning of expectation.

(b) [3] Let $Y = 2X + 1$. Compute $E[Y]$ using linearity of expectation. Check the result by calculating $E[Y]$ directly from the possible values of Y and their probabilities.

Solution (a). The probabilities, rather than the number of distinct values, supply the weights:

$$E[X] = (-3)\frac{1}{4} + 1\frac{1}{2} + 4\frac{1}{4} = \frac{3}{4}.$$

The support is $\{-3, 1, 4\}$, so $3/4$ is not a possible realization. A probability-weighted average need not be a support value; it summarizes the distribution. Here it lies between -3 and 4.

Solution (b). Linearity gives $E[Y] = 2E[X] + 1 = 2(3/4) + 1 = 5/2$. The transformed values are $-5, 3, 9$, with masses $1/4, 1/2, 1/4$. Direct averaging gives

$$(-5)\frac{1}{4} + 3\frac{1}{2} + 9\frac{1}{4} = \frac{-5 + 6 + 9}{4} = \frac{5}{2}.$$

No independence assumption is needed for an affine change of one variable.

2. Drawing colored tokens

5 points

Draw four tokens in order, without replacement, from ten distinct tokens: four red and six blue. At each draw, every remaining token is equally likely to be selected. Let I_j equal 1 if draw j is red and 0 otherwise, for $j = 1, 2, 3, 4$. Let R be the number of red tokens among the four draws.

(a) [3] Express R in terms of I_1, I_2, I_3, I_4. Find $E[I_j]$ for each draw and use these expectations to find $E[R]$. State whether this calculation requires the indicators to be independent.

(b) [2] Find $P(I_2 = 1 \mid I_1 = 1)$ and $P(I_2 = 1)$. Use the comparison to decide whether I_1 and I_2 are independent. Does your conclusion affect the validity of part (a)? Explain.

Solution (a). The count is $R = I_1 + I_2 + I_3 + I_4$. By symmetry each draw has unconditional red probability $4/10 = 2/5$. The indicator identity and linearity give

$$E[I_j] = \frac{2}{5} \quad (j = 1, 2, 3, 4), \qquad E[R] = \sum_{j=1}^{4} E[I_j] = \frac{8}{5}.$$

Linearity of expectation holds for a finite sum whether or not the variables are independent.

Solution (b). After a red first draw, three red tokens remain among nine:

$$P(I_2 = 1 \mid I_1 = 1) = \frac{3}{9} = \frac{1}{3} \neq \frac{2}{5} = P(I_2 = 1).$$

The indicators are dependent. Part (a) remains valid because it used unconditional indicator means and linearity, neither of which required independence. A binomial assumption is not justified by this mechanism.

3. The square of a discrete random variable

5 points

Each value in $\{-2, -1, 0, 1, 2\}$ has probability $1/5$ for X. Define $Z = X^2$.

(a) [3] Find the complete PMF of Z and then compute $E[Z]$ from that PMF.

(b) [2] Compute $E[X]$ and $E[X^2]$ directly from the PMF of X. Is $E[X^2] = (E[X])^2$ true for this distribution? Justify your answer.

Solution (a). Only the values $0, 1, 4$ occur for Z. The input 0 gives $Z = 0$; inputs $-1, 1$ give $Z = 1$; inputs $-2, 2$ give $Z = 4$. Hence

$$p_Z(z) = \frac{1}{5} I_{\{z=0\}} + \frac{2}{5} I_{\{z=1\}} + \frac{2}{5} I_{\{z=4\}}, \qquad E[Z] = 0\frac{1}{5} + 1\frac{2}{5} + 4\frac{2}{5} = 2.$$

The masses are zero elsewhere and sum to one.

Solution (b). The original weights remain attached to the inputs:

$$E[X] = \frac{-2 - 1 + 0 + 1 + 2}{5} = 0, \qquad E[X^2] = \frac{4 + 1 + 0 + 1 + 4}{5} = 2.$$

Thus $E[X^2] = 2$ but $(E[X])^2 = 0$. Squaring is nonlinear, so the affine expectation rule does not justify the proposed equality. The direct second moment agrees with $E[Z]$.

4. Mean and variability of a delay

10 points

A task has additional delay D, measured in milliseconds, with

d	0	2	6
$p_D(d)$	1/2	1/3	1/6

The PMF is zero elsewhere.

(a) [5] Compute $E[D]$, $E[D^2]$, $\mathrm{Var}(D)$, and $\mathrm{SD}(D)$, using the second-moment formula for variance. State the units of all four quantities.

(b) [5] Check your variance by evaluating $E[(D - E[D])^2]$ directly from the PMF. Explain whether the variance is positive. Also evaluate $E[D - E[D]]$ and explain whether it can measure variability.

Solution (a). The weighted sums give

$$E[D] = 2\frac{1}{3} + 6\frac{1}{6} = \frac{5}{3} \text{ ms}, \qquad E[D^2] = 4\frac{1}{3} + 36\frac{1}{6} = \frac{22}{3} \text{ ms}^2.$$

Therefore

$$\mathrm{Var}(D) = \frac{22}{3} - \left(\frac{5}{3}\right)^2 = \frac{41}{9} \text{ ms}^2, \qquad \mathrm{SD}(D) = \frac{\sqrt{41}}{3} \text{ ms}.$$

The second moment squares the values before averaging; the squared mean is a different quantity.

Solution (b). The mean $5/3$ is a fixed number. The centered-square sum is

$$\frac{1}{2}\left(0 - \frac{5}{3}\right)^2 + \frac{1}{3}\left(2 - \frac{5}{3}\right)^2 + \frac{1}{6}\left(6 - \frac{5}{3}\right)^2 = \frac{25}{18} + \frac{1}{27} + \frac{169}{54} = \frac{41}{9}.$$

Every term is nonnegative, and at least one positive-probability value differs from the mean, making the variance positive. Signed deviations cancel instead:

$$E[D - E[D]] = E[D] - E[D] = 0.$$

That value would be zero even for a variable with substantial spread.

5. Recalibrating a measurement

5 points

A measurement X has mean 12 and variance 4. Define the recalibrated reading $Y = 50 - 3X$.

(a) [3] Find $E[Y]$, $\mathrm{Var}(Y)$, and $\mathrm{SD}(Y)$. Explain how the coefficient -3 affects the variance and standard deviation.

(b) [2] A student claims $\mathrm{Var}(50 - 3X) = 50 - 3\,\mathrm{Var}(X)$. State the correct general rule for $\mathrm{Var}(aX + b)$, where a, b are constants. If Y is replaced by $Y^* = 55 - 3X$, which quantities in part (a) change? Justify your answer by comparing $Y - E[Y]$ with $Y^* - E[Y^*]$.

Solution (a). Affine linearity and variance scaling give

$$E[Y] = 50 - 3(12) = 14, \qquad \mathrm{Var}(Y) = (-3)^2(4) = 36, \qquad \mathrm{SD}(Y) = 6.$$

Indeed $Y - E[Y] = -3(X - 12)$, so squared deviations are multiplied by 9. Taking the nonnegative square root multiplies SD by $|-3| = 3$, rather than by -3.

Solution (b). Variance is not linear: $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$. Let $Y' = 55 - 3X$. Its mean is $55 - 36 = 19$, while

$$Y' - E[Y'] = 55 - 3X - 19 = -3(X - 12).$$

The centered deviations are the same as for Y, so variance remains 36 and SD remains 6. Changing the constant shifts location without changing spread.

6. Mean and variance of an indicator

5 points

Let I indicate an event with probability p, where $0 \leq p \leq 1$. Thus $P(I = 1) = p$ and $P(I = 0) = 1 - p$.

(a) [3] Compute $E[I]$ and $E[I^2]$ from the two possible values of I and their probabilities. Use these results to derive $\text{Var}(I)$.

(b) [2] Calculate the variance when $p = 1/5$ and when $p = 1/2$. Which has greater variability, as measured by variance? Find the variance for $p = 0$ and for $p = 1$, and explain these two cases.

Solution (a). Both $0^2 = 0$ and $1^2 = 1$, hence $I^2 = I$. Direct averaging gives $E[I] = 0(1 - p) + 1p = p$, and therefore $E[I^2] = p$. Consequently

$$\text{Var}(I) = E[I^2] - (E[I])^2 = p - p^2 = p(1 - p).$$

Solution (b). For $p = 1/5$, variance is $(1/5)(4/5) = 4/25$. For $p = 1/2$, it is $1/4$. Since $4/25 < 1/4$, the second indicator has greater variance (and also greater SD). At $p = 0$ the indicator is always zero; at $p = 1$ it is always one. Both are constant, so there is no deviation from the mean and the variance is zero.

7. A joint PMF and the sum of two variables

10 points

The joint PMF of X, Y is given by the table below and is zero for all other pairs. The constant c makes it a valid PMF.

$X\backslash Y$	0	1
0	c	$2c$
1	$2c$	c
2	0	$4c$

(a) [5] Find c and the complete marginal PMFs $p_X(x)$ and $p_Y(y)$. Verify that the joint probabilities sum to 1 and that each marginal PMF also sums to 1.

(b) [5] Define $S = X + Y$. Find the complete PMF of S and compute $E[S]$ from it. Check the result by calculating $E[X] + E[Y]$ from the marginal PMFs.

Solution (a). The table has total mass $(1+2+2+1+0+4)c = 10c$, so $c = 1/10$. Its complete normalized form is

$X\backslash Y$	0	1	$p_X(x)$
0	1/10	1/5	3/10
1	1/5	1/10	3/10
2	0	2/5	2/5
$p_Y(y)$	3/10	7/10	1

Thus p_X has masses $3/10, 3/10, 2/5$ on $\{0, 1, 2\}$, and p_Y has masses $3/10, 7/10$ on $\{0, 1\}$, with zero elsewhere. All entries are nonnegative. Joint, row-total, and column-total masses each sum to one.

Solution (b). Group by $s = x + y$, including the zero cell without assigning it any mass:

s	cells (x, y)	$p_S(s)$
0	$(0, 0)$	1/10
1	$(0, 1), (1, 0)$	2/5
2	$(1, 1), (2, 0)$	1/10
3	$(2, 1)$	2/5

The PMF is zero elsewhere and these four masses sum to one. It gives

$$E[S] = 0\frac{1}{10} + 1\frac{2}{5} + 2\frac{1}{10} + 3\frac{2}{5} = \frac{9}{5}.$$

The marginals give $E[X] = 1(3/10) + 2(2/5) = 11/10$ and $E[Y] = 7/10$. Their sum is $18/10 = 9/5$. This linearity check does not require independence.

8. Conditional probabilities from a joint PMF

5 points

Use the joint PMF in Question 7.

(a) [3] Find the complete conditional PMF $p_{Y|X}(y \mid 1)$ and the probability $P(X = 1 \mid Y = 1)$. Show the conditional-probability calculation for each.

(b) [2] Find the complete conditional PMF $p_{X|Y}(x \mid 0)$ and verify that it sums to 1. Is $P(Y = 1 \mid X = 3)$ defined by the elementary definition of conditional probability? Explain.

Solution (a). The row $X = 1$ has mass $3/10 > 0$. Divide its entries by that row total:

$$p_{Y|X}(y \mid 1) = \frac{2}{3} I_{\{y=0\}} + \frac{1}{3} I_{\{y=1\}}.$$

For the reverse question the denominator is the column total $P(Y = 1) = 7/10$:

$$P(X = 1 \mid Y = 1) = \frac{p_{X,Y}(1, 1)}{p_Y(1)} = \frac{1/10}{7/10} = \frac{1}{7}.$$

The two calculations condition on different events.

Solution (b). Column $Y = 0$ has total mass $3/10 > 0$. Its normalized entries at $x = 0, 1, 2$ are $1/3, 2/3, 0$, respectively. Equivalently,

$$p_{X|Y}(x \mid 0) = \frac{1}{3} I_{\{x=0\}} + \frac{2}{3} I_{\{x=1\}}.$$

This PMF is zero elsewhere and sums to one; its positive-mass support is $\{0, 1\}$. Since $P(X = 3) = 0$, the elementary ratio for $P(Y = 1 \mid X = 3)$ has a zero denominator and does not define that conditional probability. Assigning it zero by dividing zero by zero is not valid.

9. Comparing two joint distributions

5 points

Use the marginal PMFs from Question 7. Let U and V be independent, with $p_U(u) = p_X(u)$ for every u and $p_V(v) = p_Y(v)$ for every v. Thus U has the same marginal distribution as X, and V has the same marginal distribution as Y.

(a) [3] Are the original variables X and Y independent? Justify your answer with a probability comparison. Construct the complete joint PMF table for U and V.

(b) [2] Calculate and compare $P(X + Y = 3)$ and $P(U + V = 3)$. Then calculate and compare $E[X + Y]$ and $E[U + V]$. Explain what the two comparisons show about the role of the joint distribution.

Solution (a). The zero cell is decisive:

$$p_{X,Y}(2, 0) = 0 \neq p_X(2)p_Y(0) = \frac{2}{5}\frac{3}{10} = \frac{3}{25}.$$

Both marginal values have positive probability, so this refutes independence. Under independence every new cell is a marginal product:

$U\backslash V$	0	1	$p_U(u)$
0	9/100	21/100	3/10
1	9/100	21/100	3/10
2	12/100	28/100	2/5
$p_V(v)$	3/10	7/10	1

The PMF is zero elsewhere. Row and column totals reproduce the specified marginals; all cells factor, establishing independence.

Solution (b). The only pair summing to three is $(2, 1)$ in either model. Hence

$$P(X + Y = 3) = \frac{2}{5}, \qquad P(U + V = 3) = \frac{2}{5}\frac{7}{10} = \frac{7}{25}.$$

The means agree by linearity and the equal marginals:

$$E[X + Y] = E[U + V] = \frac{11}{10} + \frac{7}{10} = \frac{9}{5}.$$

Marginals determine a sum's mean through linearity, but they do not determine all probabilities involving both variables. Those probabilities depend on the joint law.

10. A joint PMF from conditional probabilities

5 points

The random variable A takes values 0 and 1, with $P(A = 0) = 2/5$ and $P(A = 1) = 3/5$. For each possible value of A, the conditional PMF of B is given below; it is zero for all other b:

b	0	1	2
$p_{B\|A}(b \mid 0)$	1/2	1/4	1/4
$p_{B\|A}(b \mid 1)$	1/6	1/2	1/3

(a) [3] Find the complete joint PMF $p_{A,B}(a, b)$ and the complete marginal PMF $p_B(b)$. Verify that each PMF sums to 1.

(b) [2] Find $P(A = 0 \mid B = 2)$. Are A and B independent? Justify your conclusion with a probability comparison.

Solution (a). Multiply each conditional row by its own conditioning probability:

$$p_{A,B}(a, b) = p_A(a)p_{B|A}(b \mid a).$$

The complete table is

$A\backslash B$	0	1	2	$p_A(a)$
0	1/5	1/10	1/10	2/5
1	1/10	3/10	1/5	3/5
$p_B(b)$	3/10	2/5	3/10	1

Set the joint PMF to zero elsewhere. The marginal is $3/10, 2/5, 3/10$ on $\{0, 1, 2\}$ and zero elsewhere. Every entry is nonnegative. The row totals sum to one; the column totals also sum to one.

Solution (b). The new conditioning event has probability $P(B = 2) = 3/10 > 0$, so

$$P(A = 0 \mid B = 2) = \frac{p_{A,B}(0, 2)}{p_B(2)} = \frac{1/10}{3/10} = \frac{1}{3}.$$

The variables are dependent: the given conditional probability $P(B = 2 \mid A = 0) = 1/4$ differs from the marginal $P(B = 2) = 3/10$, with $P(A = 0) > 0$. Equivalently, $p_{A,B}(0, 2) = 1/10 \neq (2/5)(3/10) = 3/25$.